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MIL-H-46792A(AT)
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MILITARY SPECIFICATION

HEATER, VEHICULAR COMPARTMENT: MULTIFUEL BURNING 24 VOLT, DC 30,000 BTU/HR

This limited coordination military specification has been prepared by the US Army Tank-Automotive Command based upon currently available technical information but has not been approved for promulgation as a coordinated revision of MIL-H-46792A. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, testing, manufacturing, and acceptance requirements for a forced air inlet personnel heater assembly.

1.2 Item definition. The personnel heater, referred to herein as "heater", shall be a diesel fuel burning unit with an electrical control system and an electric blower.

1.2.1 Dimensions. The heater shall conform to the dimensions shown on specification sheet MIL-H-46792/2C. The arrangement of exhaust, inlet, and outlet connections shall be as shown on the specification sheet (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2540

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

GG-P-455	- Plates and Foils, Photographic (Photo-sensitive Anodized Aluminum).
QQ-P-35	- Passivation Treatments for Corrosion Resisting Steel.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
TT-E-485	- Enamel, Semi-gloss, Rust-inhibiting.
TT-E-529	- Enamel, Alkyd, Semi-gloss.
TT-P-636	- Primer Coating, Alkyd, Wood and Ferrous Metal.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-P-14105	- Paint, Heat-resisting, (for Steel Surfaces).
MIL-P-19834	- Plate, Identification, Metal Foil, Adhesive Backed.

STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment.

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MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration System Requirements.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

19.5.4	- Power Test Code.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 E. 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-2156	- Smoke Density in Flue Gases from Burning Distillate Fuels, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

210-74	- Test Code for Air Moving Devices.

(Application for copies should be addressed to the Air Movement and Control Association, 320 W. University Drive, Arlington Heights, IL 60004.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish heaters which shall be subjected to first article inspection (see 4.4 and 6.2). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent heaters delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein, in applicable specifications, or drawings. Materials not specifically designated shall be suitable for use in heaters operating over specified ranges, without any change in properties that would result in operation of the units falling outside of the specified limits (see 6.4). The heater shall be fabricated from suitable corrosion-resistant materials or treated to prevent corrosion.

3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, the compatibility of dissimilar metals shall be in accordance with MIL-STD-889.

3.3 Design and construction. The heaters shall be constructed in accordance with the requirements of MIL-H-46792/2C and shall be completely interchangeable with each other as a complete unit.

3.3.1 Weight. Weight of the heater shall not exceed 21 pounds.

3.3.2 Interchangeability. The assembly shall be so designed and fabricated that all component assemblies and parts are functionally and dimensionally interchangeable with like component assemblies and parts previously furnished by the same manufacturer. Construction shall be such that any part, except those furnished in matched sets or for which a selective fit is specified, may be installed, replaced, and adjusted without requiring modification.

3.4 Performance. The heater shall meet the performance requirements specified herein when mounted in any position from horizontal to vertical with the air inlet level with, or higher than, the air outlet (see 4.6).

3.4.1 Voltage. The heater shall start and meet performance requirements specified herein, in low and high heat, with an applied voltage of 19 to 30 volts direct current (Vdc), 24 Vdc nominal (see 4.6.1).

3.4.2 Current. At 24 Vdc, maximum current drawing from the power source shall be as follows (see 4.6.2):

Ambients above 45°F	Starting	12.0 amperes (amps)
	Running	8.0 amps
Ambients below 45°F	Starting	14.0 amps
	Running	10.0 amps

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3.4.3 Fuel. The heater shall perform as specified herein using fuel conforming to VV-F-800 as follows (see 4.6.3):

Above plus 20°F	type DF-2
Plus 20° to minus 25°F	type DF-1
Below minus 25°F	type DF-A

3.4.4 Fuel consumption. With any fuel pressure between 3 and 15 pounds per square inch gage (psig) at the fuel valve inlet, fuel consumption shall be as follows (see 4.6.4):

High heat	0.049 pounds per minute (lbs/min) maximum
Low heat	0.031 lbs/min maximum

3.4.5 Heat output. With a ventilating back pressure of .75 inch of water and an exhaust back pressure of 0.75 inch of water, heater output shall be as follows (see 4.6.5):

High heat	19.0 to 30.0 Vdc, 27,000 British Thermal Units per hour (BTU/HR) minimum
Low heat	19.0 to 30.0 Vdc, 15,000 BTU/HR ± 20 percent

The air flow restrictions specified are to be set at 24 Vdc input only and maintained in this position at all voltages.

3.4.6 Air discharge rate. At low heat and high heat, with 24 Vdc applied, the ventilating air discharge rate, based on standard air density of 0.075 pounds per cubic foot at 29.92 inches of mercury, 70°F, shall be as follows (see 4.6.6):

Pressure	Discharge rate
At 0.00 inch (in.) water ventilating back pressure nominal and 0.00 in. water exhaust back pressure nominal.	115 standard cubic feet per minute (scfm), minimum.
At .75 in. water ventilating back pressure nominal, and 0.75 in. water exhaust back pressure nominal.	80 scfm, minimum.

3.4.7 Overheat switch. The heater shall have an overheat switch which will automatically open a circuit and shut off fuel flow when the temperature of the discharge air reaches 400° plus or minus 75°F (see 4.6.7).

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3.4.8 Smoke density. After the heater has been operating for five minutes, minimum, exhaust gases shall not exceed a smoke spot number of three, except that at an ambient temperature of 73° ± 18°, and a voltage of 19 ± 0.5 Vdc, the smoke spot number shall not exceed five. Smoke spot numbers are defined in ASTM D-2156-65 (see 4.6.8).

3.4.9 Starting time. The starting time, measured from the time the heater is turned on until the igniter circuit is de-energized, shall not be greater than 4.0 minutes (see 4.6.9).

3.4.10 Purging time. The purging time, measured from the time the heater switch is turned off until the blowers shut off, shall not be greater than 3.5 minutes (see 4.6.10).

3.4.11 Fuel leakage. There shall be no visible leakage from the heater fuel components during a period of five minutes, minimum, with the heater operating and 15 psig fuel pressure applied to the inlet (see 4.6.11).

3.4.12 Toxic fumes. Combustion and ventilating air in the heater shall be isolated from each other, after combustion. In operation, the heater shall not contaminate the ventilating air with products of combustion (see 4.6.12).

3.4.13 Endurance. The heater shall meet the requirements of 3.4.1 through 3.4.12 after 800 hours of operation; including 2400, minimum, start cycles. The following maintenance is permissible (see 4.6.13):

a. Cleaning of fuel filters and screens.
b. Cleaning and adjustment of the flame detector switch, but not more than two times.
c. Replacment of the igniter, but not more than two times.
d. Use of a burner repair kit consisting of a fuel vaporizer and starting wick (with necessary mounting parts), but not more frequently than every 250 hours.

3.4.14 Environmental conditions. The heater shall meet the requirements of 3.4.1 through 3.4.12 after being subjected to each environmental condition specified in 3.4.14.1 through 3.4.14.6 (see 4.6.14).

3.4.14.1 Operating temperature. The heater shall start and operate as specified herein at any temperature from plus 70°F + 5°F to minus 55°F + 5°F, provided that fuel is supplied in liquid form (see 4.6.14.1 and 4.6.14.1.1).

3.4.14.2 Storage temperature. The heater shall not be damged by storage conditions reanging from minus 65°F to plus 160°F (see 4.6.14.2).

3.4.14.3 Humidity. The heater shall withstand relative humidity up to 100 percent (see 4.6.14.3).

3.4.14.4 Corrosion. The heater shall function as specified herein after exposure to salt fog for 96 hours (see 4.6.14.4).

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3.4.14.5 Shock. The heater shall function as specified herein after being subjected to 40g shock loading. Input to be measured at the interface of the heater and support bracket (see 4.6.14.5).

3.4.14.6 Vibration. The heater shall function as specified herein during and after sinusoidal vibration in accordance with figure 1, including up to four resonance dwells. Input to be measured at the interface of the heater and support bracket (see 4.6.14.6).

3.4.14.7 Electromagnetic radiation. During any mode of operation, or while changing modes of operation, the heater shall meet the requirements of MIL-STD-461, Notice 4, as modified herein by figures 2 thru 6. Emissions having a duration not exceeding one second, and recurring not more than once in three minutes, are exempt. Applicability of MIL-STD-461 and figures 2 through 6 shall be as follows (see 4.6.14.7):

Requirement (MIL-STD-461)	Limits MIL-STD-461	Limits (This specification)
CE01		Figure 2
CE04 (+28 Vdc line)		Figure 3
RE02		Figures 4 and 5

3.5 Protective coating. All steel parts, except stainless steel, that are subjected to temperatures exceeding 200°F, shall be painted with two coats of high temperature olive drab enamel conforming to MIL-P-14105. Surfaces shall be free of oil, grease, rust and mill scale before applying enamel finish. All other painted steel parts shall be furnished in accordance with MIL-P-14105, or primed with primer conforming to TT-P-636, and finished with enamel conforming to TT-E-529 or TT-E-485, color to match 24087 of FED-STD-595.

3.5.1 Steel components. Unpainted steel components, other than stainless steel, shall be cadmium plated in accordance with QQ-P-416, type II, class 2. Radiation surfaces may be aluminized steel.

3.5.2 Aluminum components. Aluminum components shall be anodized in accordance with MIL-C-5541, class 1A.

3.6 Marking and identification plates.

3.6.1 Marking. Unless otherwise specified (see 6.2), heater shall be marked in accordance with MIL-STD-130.

3.6.2 Identification plates. As specified (see 6.2), nameplates shall be furnished in one of the following configurations:

a. Material and construction shall conform to MIL-P-514, type III, composition A, class 2, or composition C. Composition C material shall conform to GG-P-455, type II, grade A, class 1.

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b. Adhesive backed aluminum foil shall conform to MIL-P-19834, type I, glossy: except backing sheet, which may be of paper, shall be removable intact by stripping without use of water or other solvents. Plate shall have an average adhesion, over bare and painted metal surfaces, of not less than 50 ounces per inch of width at 77° ± 5°F.

3.7 Workmanship. Heater shall be free of defects such as burrs, scratches, chips sharp edges, corrosion, scale and dirt. Workmanship shall also be in accordance with requirement 9 of MIL-STD-454.

3.8 Safety. Hot spots that could create a hazard to personnel shall be minimized and shall be identified on the drawings by the contractor. The heater shall be so designed that, with proper installation, heater exhaust gases cannot enter the vehicle personnel compartment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract, the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designed inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Materials and construction. To determine conformance to 3.2 through 3.3.2, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Acceptance tests (see 4.5.3.1).
   3. Control tests (see 4.5.3.2).

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4.3 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions:

a.	Air temperature	73° ± 18°F
b.	Barometric pressure	28.5 + 2.0 inches of mercury - 4.5
c.	Relative humidity	50 ± 30 percent
d.	Fuel	Diesel grade DF-2 (see 3.4.3)
e.	Fuel pressure	8 ± 1 psi
f.	Mounting position	As specified (see table II)

4.4 First article inspection. First article inspections are categorized as preproduction inspection and initial production inspection. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply heaters that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of three heaters. Preproduction inspection samples shall be examined per table I and inspected per table II under the surveillance of the Government inspector.

4.4.2 Initial production inspection. During fabrication of the first production heater, an in-process inspection shall be conducted by the contractor under the surveillance of the Government inspector. Inspection shall be made at the contractor's facility prior to application of primer and paint. Processing and welding procedures, quality system and inspection records shall be reviewed and evaluated during this inspection.

4.4.2.1 Test routine. Following in-process inspection, three heaters that have been produced with production materials, equipment and procedures shall be subjected to inspection by the contractor under the surveillance of the Government inspector. Initial production units shall be examined per table I and inspected per table II.

4.4.3 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval until corrective action by the contractor has been approved by the Government.

4.4.4 Heater disposition. After completion of inspection, the first production heater shall remain at the manufacturing facility, as a production sample, and be the last heater shipped on the contract. The heater may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the heater during this period.

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4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the heaters of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105. Before sampling may be initiated, the contractor shall 100 percent inspect the first 100 heaters to establish a process average (see 6.3) to allow normal sampling in accordance with MIL-STD-105.

4.5.2 Quality conformance examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table I.

TABLE I. Classification of defects.

Category	Defect	Method of examination
Major		
101	Improper materials (see 3.2).	Visual
102	Dimensions affecting interchangeability not within tolerance (see 3.3).	SIE 1/
103	Safety defects (see 3.8).	Visual, tactile
Minor		
201	Weight over maximum allowed (see 3.3.1).	Scale
202	Improper protective coating (see 3.5).	Visual
203	Improper marking or identification (see 3.6).	Visual
204	Faulty workmanship (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent). Each heater shall be subjected to the tests specified in table II.

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4.5.3.2 Control tests. Control tests shall be conducted on one heater of each lot of 600 units consecutively produced, except that not more than one unit shall be tested in a three month period, nor less than one unit in a six month period. The heaters shall be examined per table I and tested per table II.

4.5.4 Failure. Failure of any heater to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Conformance verification. Except as otherwise specified, performance tests shall be conducted with heater mounted in position specified in table II, at both high and low heat, at input voltages of 19.0 ± 0.5 Vdc, 24.0 ± 0.5 Vdc and 30.0 ± 0.5 Vdc. Acceptance inspection shall be conducted at 24.0 ± 0.5 Vdc only. Prior to each start, heater shall be cooled for a minimum of 10 minutes.

4.6.1 Voltage test. Applied voltage shall be monitored throughout performance testing to determine conformance to 3.4.1.

4.6.2 Current test. Current draw from the power source shall be monitored during performance testing, while starting and running at ambient temperatures above and below 45°F, to determine conformance to 3.4.2. Running current shall be measured after the igniter is de-energized.

4.6.3 Fuel test. Acceptable performance during testing as specified in 4.6.13 and 4.6.13.1, using the fuel types specified, shall determine conformance to 3.4.3.

4.6.4 Fuel consumption test. Fuel flow shall be measured with a fuel pressure of 15 (+0, -1) psig applied to the fuel valve inlet and subsequently with a pressure of 3 (+1, -0) psig, similarly applied, to determine conformance to 3.4.4.

4.6.5 Heat output test. With a ventilating back pressure of 0.75 inch of water and an exhaust back pressure of 0.75 inch of water (both established at 24 Vdc input only), thermocouples measuring input and output air temperatures; and an air flow meter measuring output air flow, the heater shall be operated until the outlet temperature has stabilized. Output thermocouples shall be located 42 ± 1 inches from the heater, within the duct used in 4.6.6. When testing is performed at ambient temperatures above 60°F, and the overheat switch cycles, the switch may be bypassed. Bypassing is only allowed when outlet temperature exceeds 325°F and not more than 475°F. Heat output shall be calculated, using the following equation, to determine conformance to 3.4.5.

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TABLE II. Classification of inspection.

Title	Requirement	Test	First article and first production Samples 1	2	3	Acceptance tests (100%)	Control tests
			Horiz	Vert	Horiz	Horiz	Horiz
Performance	3.4	4.6	X	X	X	X	
Voltage	3.4.1	4.6.1	X	X	X	X	
Current	3.4.2	4.6.2	X	X	X	X	
Fuel	3.4.3	4.6.3	X	X	X	X	
Fuel consumption	3.4.4	4.6.4	X	X	X	X	
Heat output	3.4.5	4.6.5	X	X	X	X	
Air discharge rate	3.4.6	4.6.6	X	X	X	X	
Overheat switch	3.4.7	4.6.7	X	X	X	X	
Smoke density	3.4.8	4.6.8	X	X	X	X	
Starting time	3.4.9	4.6.9	X	X	X	X	
Purging time	3.4.10	4.6.10	X	X	X	X	
Fuel leakage	3.4.11	4.6.11	X	X	X	X	
Toxic fumes	3.4.12	4.6.12	X	X	X	X	
Endurance	3.4.13	4.6.13	X	X			
Endurance control		4.6.13.1					X
Low temperature operation	3.4.14.1	4.6.14.1	X	X	X		X
High temperature operation		4.6.14.1.1	X	X	X		
Storage temperature	3.4.14.2	4.6.14.2	X		X		
Humidity	3.4.14.3	4.6.14.3		X			
Corrosion	3.4.14.4	4.6.14.4	X				
Shock	3.4.14.5	4.6.14.5	X	X			X
Vibration	3.4.14.6	4.6.14.6		X	X		X
Electromagnetic radiation	3.4.14.7	4.6.14.7			X		

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$$H = V \times d \times CP \times T$$

H = Heat output in BTU/hour
V = Air flow in cu. ft. per hour (Standard air)
d = Density of air at 70°F, 0.0750 lbs. per cu. ft. (standard air)
Cp = Specific heat of air, 0.24 BTU per °F, as corrected for moisture content
T = (outlet temperature, in °F) minus (inlet temperature, in °F)

4.6.6 Air discharge rate test. To determine conformance to 3.4.6, measure the ventilating air discharge velocity with 24 Vdc input, and back pressure as specified, utilizing AMCA Standard 210-67, (pitot traverse in outlet duct, or a venturi complying with ASME Power Test Code 19.5.4-1959). The outlet duct shall be insulated with a minimum of 1.5 inches of fiberglass. The air discharge rate shall be calculated using the methods defined in the AMCA Standard for 24 Vdc.

4.6.7 Overheat switch test. To determine conformance to 3.4.7, operate the heater at high heat into the duct used in 4.6.6, restricting the air flow to cause a temperature buildup in the duct. Measure the temperature of the ventilating air at a point 42 ± 1 inches from the heater, within the duct. The overheat switch shall break the circuit and cut off the fuel when the air temperature in the duct exceeds 325°F but is less than 475°F.

NOTE: Heater radiation effects should be negated by appropriate shielding of heat sensing devices.

4.6.8 Smoke density test. To determine conformance to 3.4.8, using the test methods set forth in ASTM D-2156-65, measure the smoke density of the heater exhaust gases within 6 ± 1 minutes after ignition. The smoke density shall not exceed a smoke spot number of three (3) above 24 Vdc, and a smoke spot number of five (5) at 19 ± 0.5 Vdc.

4.6.9 Starting time test. To determine conformance to 3.4.9, measure and record the time from the starting switch being turned on until the igniter is de-energized. The elapsed time shall not exceed 4.0 minutes. This test shall be conducted at low heat only.

4.6.10 Purging time test. To determine conformance to 3.4.10, after test of 4.6.9, turn the temperature control to high heat. Turn off the starting switch. Measure the time from the starting switch being turned off until the blowers shut off. The elapsed time shall not exceed 3.5 minutes.

4.6.11 Fuel leakage test. The heater shall be operated for a minimum of 5 minutes at a fuel pressure of 14.0 ± 1 psi. The heater shall subsequently be examined for evidence of fuel leakage to determine conformance to 3.4.11.

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4.6.12 Toxic fumes test. To determine conformance to 3.4.12, for heaters with a separate combustion air inlet, the inlet shall be sealed and 3 inches of water pressure applied to the exhaust outlet. Leakage shall not exceed one scfm. Prior to assembly of the heat exchanger into the heater, it shall be tested for leakage at 10 psig under water. No leakage shall be permitted. After assembly, 10 psig leakage test need not be repeated during subsequent testing.

4.6.13 Endurance test. To determine conformance to 3.4.13, the heater shall be operated for a total of 800 hours "ON" time (ignite and run) (see 4.6.5.2.1). Operation shall be in 30 minute cycles consisting of 20 minutes "ON" time and 10 minutes "OFF" time (purge and cool). The total 800 hours of operation shall consist of only "ON" time. "OFF" time will not be included in compilation of operating time. Operation shall consist of the following phases:

Phase I - Using DF fuel at room ambient (73° ± 18°F),

a. Set controls at low heat and operate heater in cycles previously described for 8 hours with an input voltage of 19.0 ± 0.5 Vdc.
b. Repeat step a using voltage input of 24.0 ± 0.5 Vdc.
c. Repeat step a using voltage input of 30.0 ± 0.5 Vdc.
d. Reset temperature control to high heat and repeat steps a, b and c.
e. Repeat steps a through d until 600 thirty minute cycles are completed (200 "ON" hours).
f. Test heater as specified in 4.6.1 through 4.6.12.

Phase II - Using DF-1 fuel at minus 25° ± 5°F (both fuel and heater) repeat steps a through f, except 300 thirty minute cycles (100 "ON" hours shall be completed.

Phase III - Using DF-A fuel at minus 55° ± 5°F (both fuel and heater) repeat steps a through f, except 300 thirty minute cycles (100 "ON" hours) shall be completed.

Phase IV - Repeat phases I through III for an additional 400 hours.

4.6.13.1 Control endurance test. For control test purposes, the "ON" time endurance testing of 4.6.13 shall be reduced to 150 hours. Test shall consist of 50 hours each of phases I, II and III. Determine conformance to 3.4.13.

4.6.14 Environmental conditions tests.

4.6.14.1 Low temperature operation test. To determine conformance to 3.4.14.1, the heater shall be subjected to the low temperature test specified in MIL-STD-810, method 502, procedure I. Temperature shall be maintained at minus 65°F for a minimum of 12 hours. Subsequently, temperature shall be raised to minus 55°F and heater tested as specified in 4.6.1 through 4.6.12 using DF-A fuel. During testing, stabilized combustion shall be maintained

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for 30 minutes on low heat and again for 30 minutes on high heat. Prior to any test, heater shall be stabilized at minus 55°F before each ignition. Subsequently, testing shall be repeated at minus 25°F using DF-1 fuel, and at plus 20°F using DF-2 fuel. At each of the three temperatures, the tests specified in 4.6.9 and 4.6.10 shall be repeated twice more, for a total of three times.

4.6.14.1.1 High temperature operation test. To determine conformance to 3.4.14.1, the heater shall be subjected to the high temperature test specified in MIL-STD-810, method 501, procedure I, except that steps 4 and 5 of the procedure shall be omitted.

4.6.14.2 Storage temperature test. To determine conformance to 3.4.14.2, heater shall be stored at 160° ± 5°F for 24 hours and, subsequently, at minus 65° ± 5°F for 24 hours. After storage, the temperature shall be raised to minus 55° ± 5°F and the heater tested as specified in 4.6.1 through 4.6.12.

4.6.14.3 Humidity test. To determine conformance to 3.4.14.3, the heater shall be tested as specified in MIL-STD-810, method 507.1, procedure II (omitting step 8). After exposure, heater shall be tested as specified in 4.6.1 through 4.6.12.

4.6.14.4 Corrosion resistance test. To determine conformance to 3.4.14.4, the heater shall be tested as specified in MIL-STD-810, method 509, procedure I. Subsequently, heater shall be tested as specified in 4.6.1 through 4.6.12.

4.6.14.5 Shock test. To determine conformance to 3.4.14.5, the heater, with no fuel applied, shall be subjected to the half sine shock pulse configuration test specified in MIL-STD-810, method 516.2, procedure I, amplitude 40g's, duration 18ms. During testing, heater shall be subjected to input power and monitored for intermittent conditions. Subsequently, heater shall be tested as specified in 4.6.1 through 4.6.12.

4.6.14.6 Vibration test. To determine conformance to 3.4.14.6, the heater shall be rigidly mounted and subjected to vibration as specified in MIL-STD-810, method 514.2, procedure VIII, using the curve shown on figure 1, at ambient temperature. During testing, heater shall be started, purged, operated at both heats, and monitored for intermittent operation. A resonant frequency search shall be conducted along each axis, with the heater operating at low heat, over a range of 5 to 500 Hz. The test schedule shall include 13 1/3 minutes dwell time at each resonance, to a maximum of four. If more than four resonances are present, the most severe shall be used. After determining resonance at low heat, the specified resonance dwell and sinusoidal cycling tests shall be conducted. Subsequently, heater shall be tested as specified in 4.6.1 through 4.6.12.

4.6.14.7 Electromagnetic radiation test. To determine conformance to 3.4.14.7, the heater shall be tested as specified in MIL-STD-462, notice 3, and figures 2 through 5.

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The forced air inlet personnel heater furnished under this specification is intended to supply warm air to the crew compartment of military vehicles.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable drawing (see 3.3).
c. Any special marking requirements (see 3.6.1).
d. Configuration of identification plates (see 3.6.2).
e. Applicable level of packaging, packing and marking (see 5.1).
f. Preproduction inspection, if required (see 4.4.1).
g. Initial production inspection, if not required (see 4.4.2).

6.3 Process average. Sampling may be initiated if the process average value for the first 100 heaters inspected is less than the AQL specified in the classification of defects for major and minor defects.

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of heaters inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for 100 consecutive heaters is less than the specified AQL (see 4.5.2.1).

6.4 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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6.6 Supersession data. This specification in lieu of MIL-H-46792A(AT), 29 October 1982 which classified the heater specified herein as Capacity B - 30,000 BTU/HR.

Custodian:
Army - AT

Review activities:
Air Force - 99
DLA - GS

Preparing activity:
Army - AT

Project No. 2540-0329

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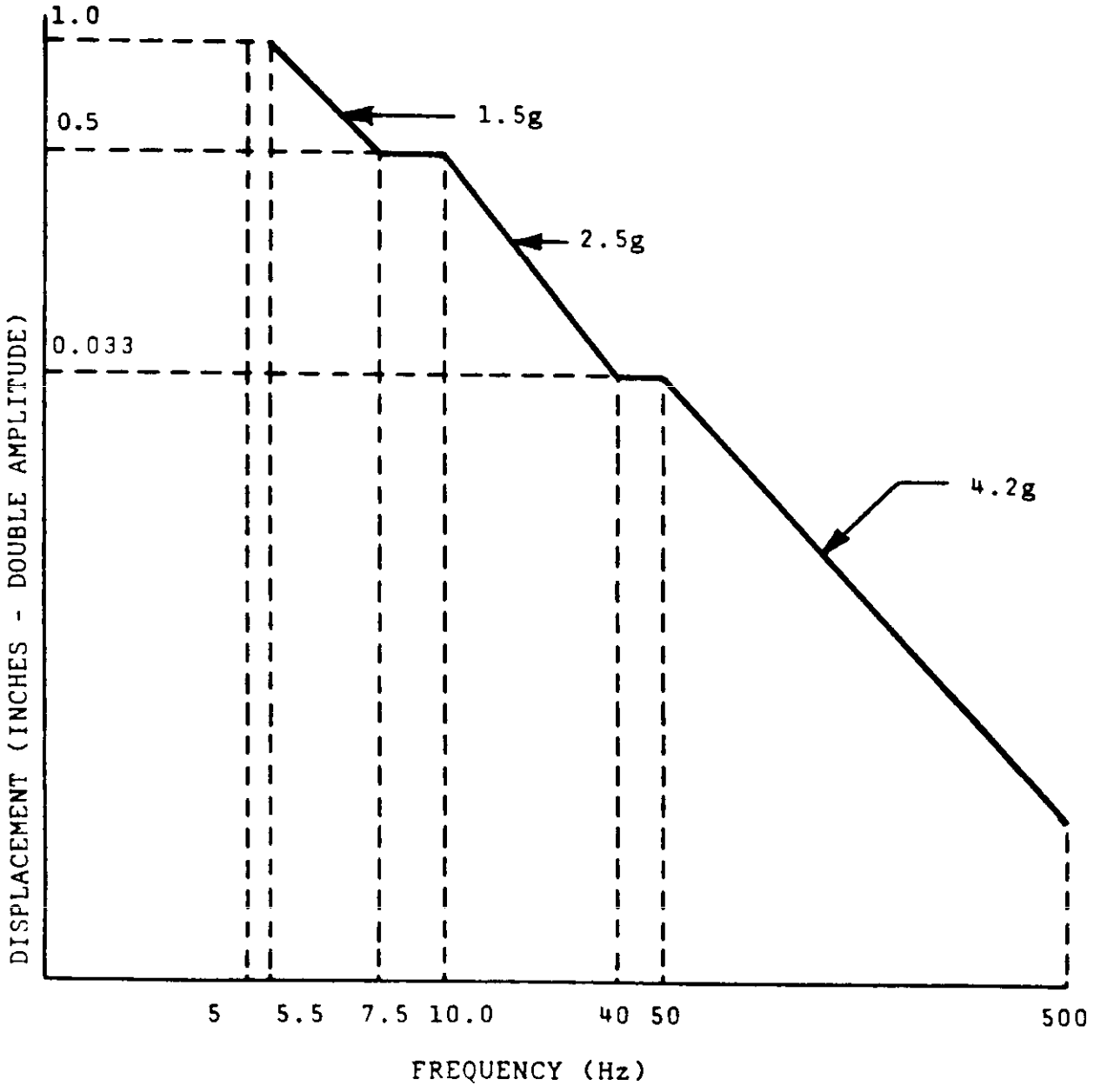


FIGURE 1. Vibration Curve.

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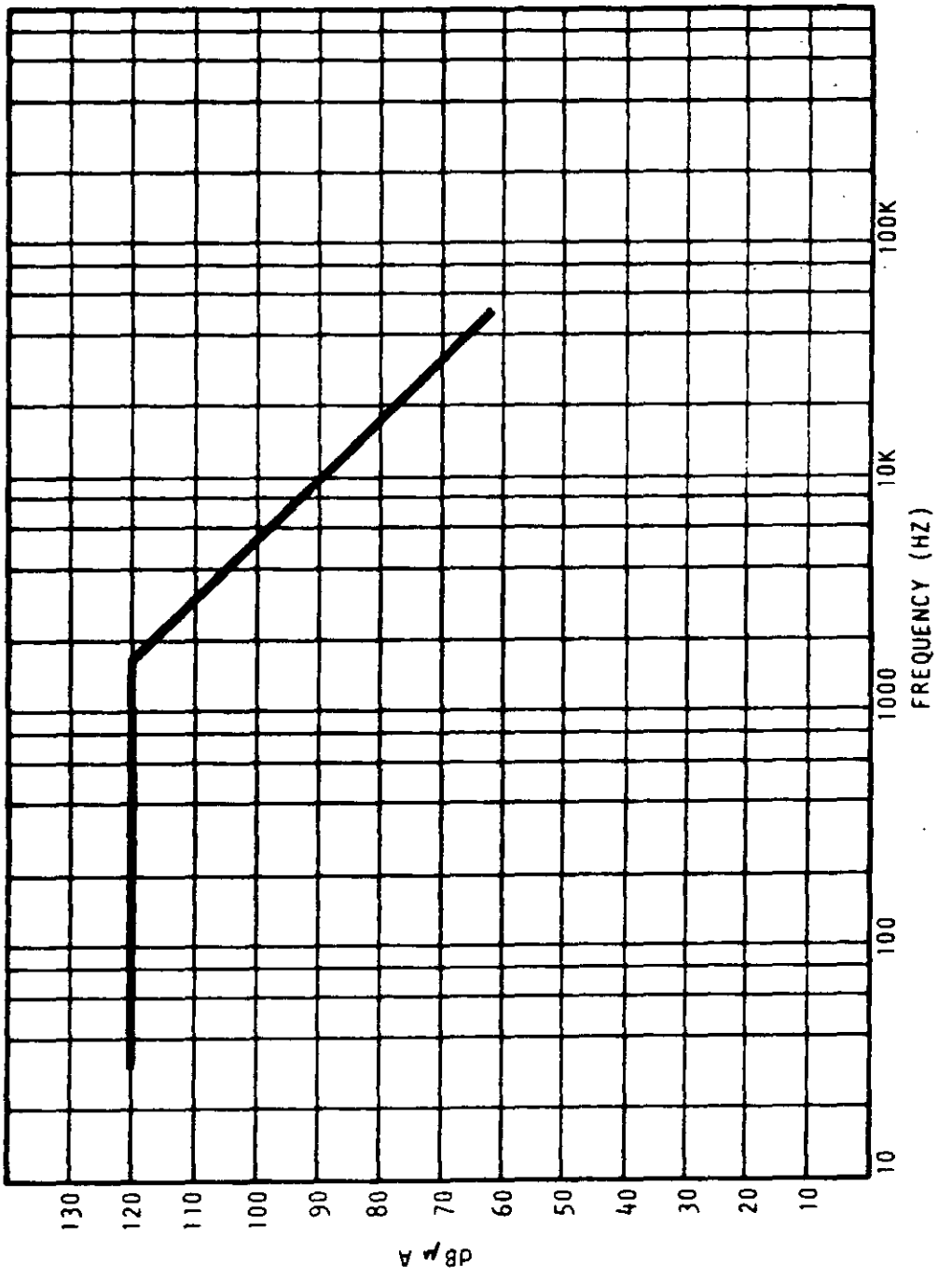


FIGURE 2. Conducted Emissions (CE01) + 28 VDC Line.

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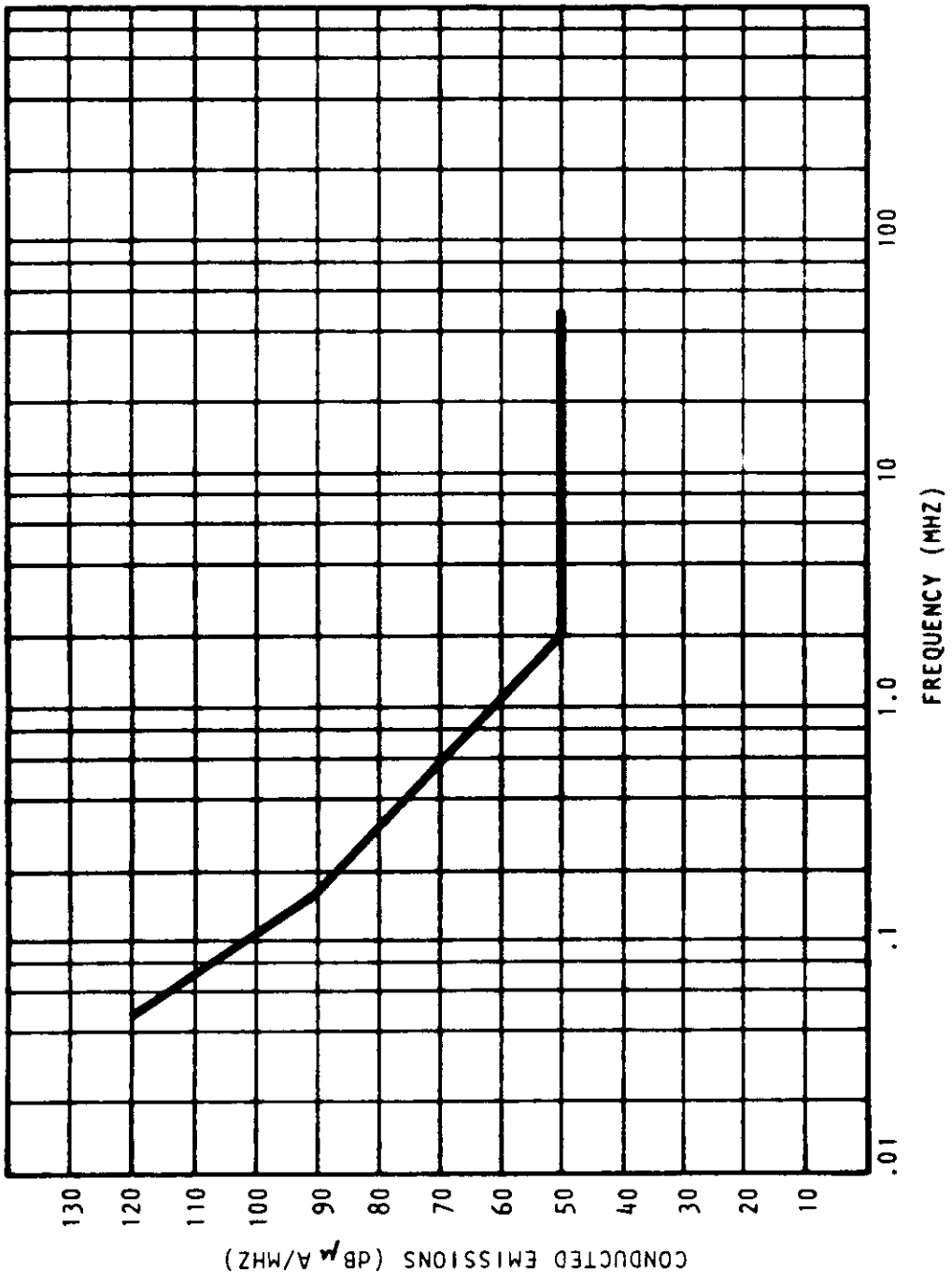


FIGURE 3. Conducted Emissions (CE04) + 28 VDC Lead.

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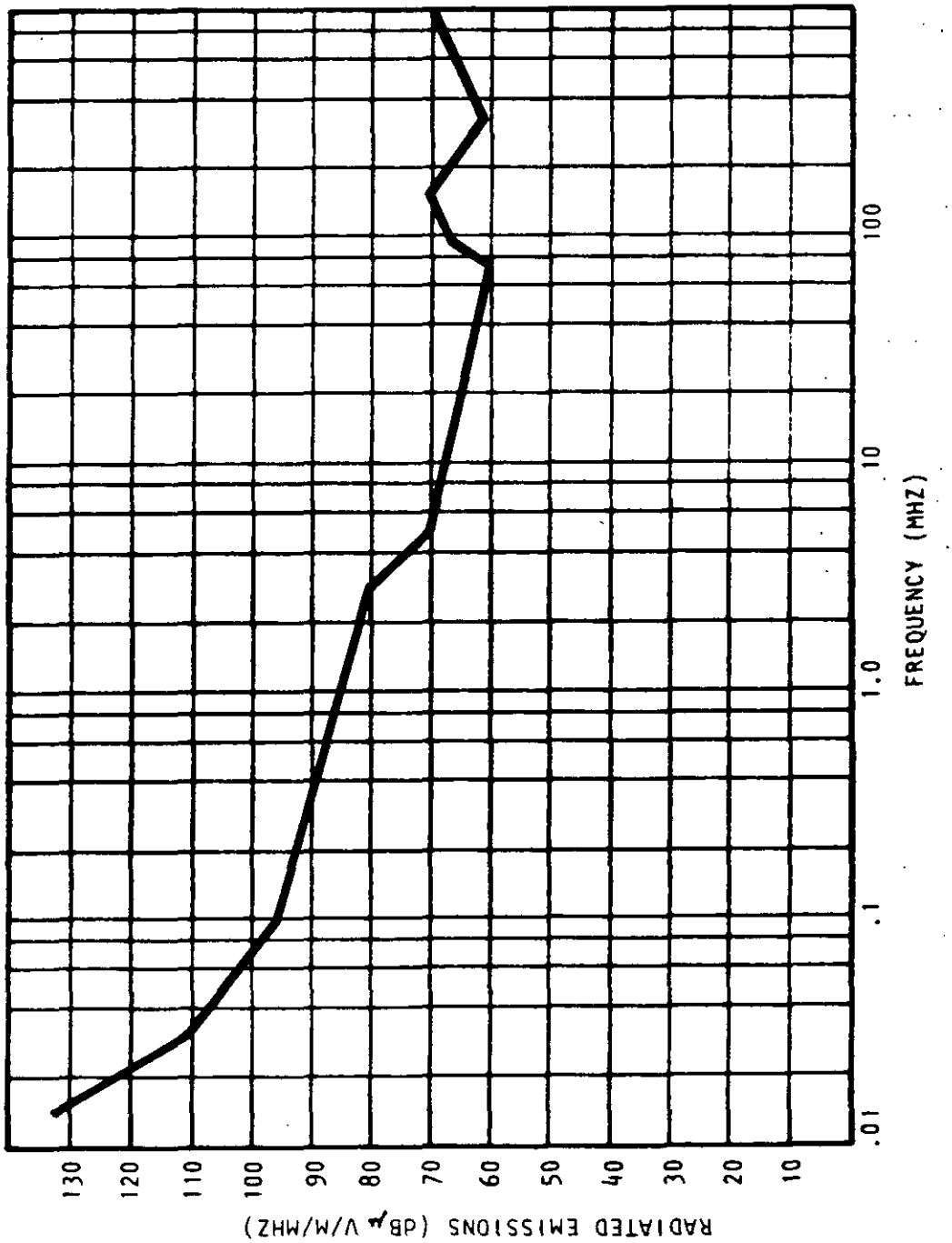


FIGURE 4. Broadband Radiated Emissions (RE02)
Flame Detect Switch - Switching Transients.

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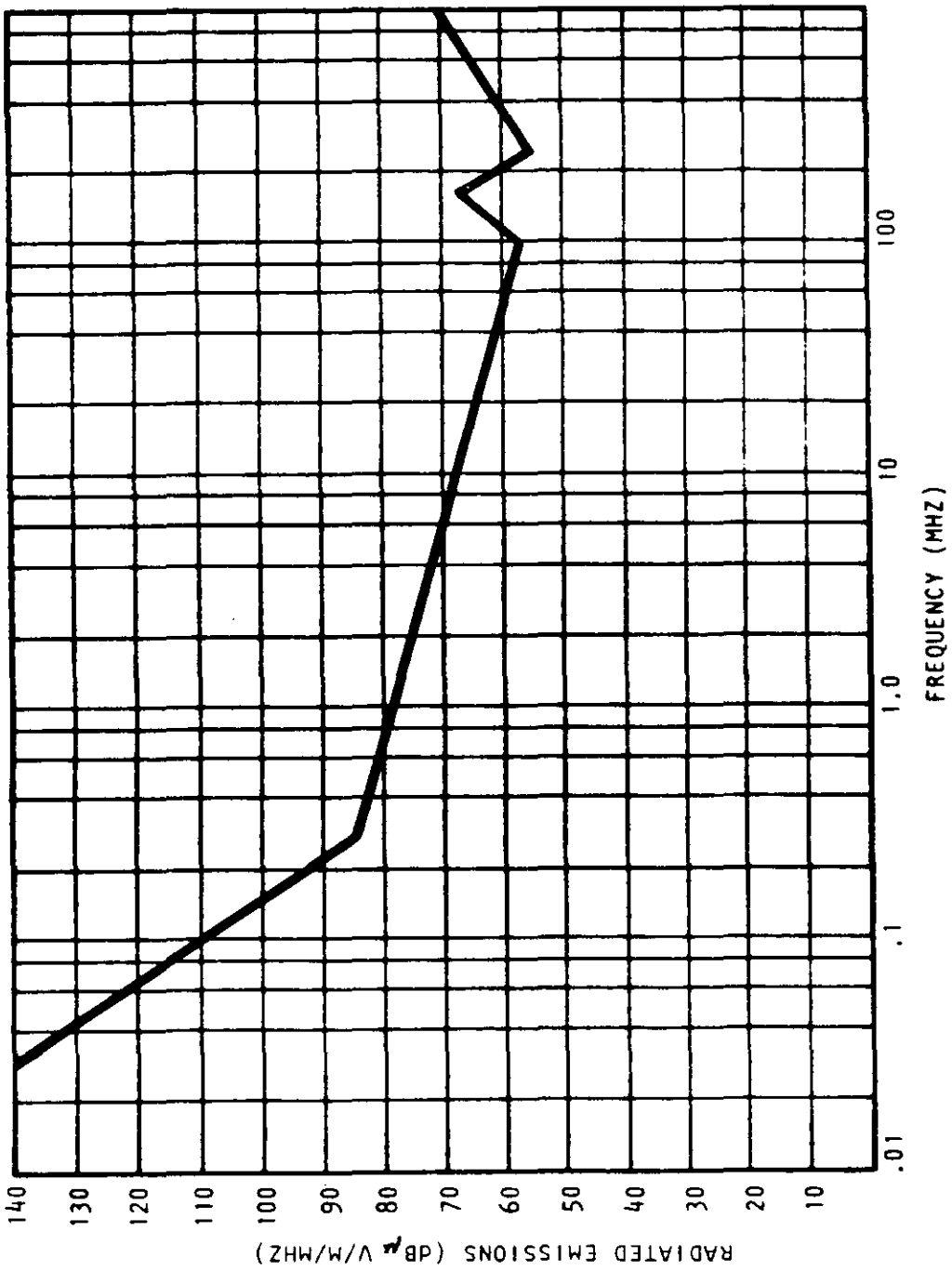


FIGURE 5. Broadband Radiated Emissions (RE02) Heater Operating - Steady State.

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	☐ VENDOR
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

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